

UC1: Place Bid

Primary Actor: Bidder

Stakeholders and Interests:

- *Bidder*: wants their bid accepted fairly and quickly, wants to know immediately if outbid
- *Seller*: wants highest possible price, wants confidence the process is tamper-proof
- *System*: must resolve concurrent bids deterministically and enforce timing rules

Preconditions:

- Bidder is logged in
- Auction is currently open (current time < auction end time)
- Item has a current highest bid (or starting price if no bids yet)

Postconditions (success):

- New bid is recorded as the current highest bid
- Bid is timestamped with millisecond precision and logged in the audit trail
- If bid was placed within the soft-close window, auction end time is extended by 5 minutes
- Previous highest bidder is notified they've been outbid
- If reserve price is met for the first time, status updates to "Reserve Met" and seller is notified

Main Success Scenario:

1. Bidder views the item detail page showing current price, minimum next bid, and live countdown.
2. Bidder enters a bid amount and submits.
3. System verifies the auction is still open (current time < end time).
4. System verifies the bidder is not already the current highest bidder.
5. System verifies the bid meets the minimum increment over the current highest bid.
6. System records the bid with a precise timestamp.
7. System updates the current highest bid and broadcasts the new price to all viewers in real time.
8. System checks if the bid falls within the soft-close window (≤ 3 minutes remaining); if so, extends the end time by 5 minutes and broadcasts the new countdown.
9. System checks if the bid meets the reserve price; if so, updates the reserve status and notifies the seller.
10. System notifies the previous highest bidder that they've been outbid.

Extensions (Alternate Flows):

- **3a. Auction has already closed:**
 1. System rejects the bid.
 2. System displays: "Sorry, this auction has already ended!"
 3. Use case ends.
- **4a. Bidder is already the current highest bidder:**
 1. System rejects the bid.
 2. System displays a message explaining they cannot outbid themselves.
 3. Use case ends.
- **5a. Bid does not meet minimum increment:**
 1. System rejects the bid.
 2. System displays the required minimum bid amount.

3. Use case ends.
- **6a. Two bids arrive concurrently for the same item:**
 1. System processes bids in strict order of arrival (by timestamp, millisecond precision).
 2. First-arriving valid bid is accepted per steps 6–10.
 3. Second bid is re-evaluated against the *new* current price (not the price the bidder saw when submitting).
 4. If the second bid is still valid (meets new minimum), it is accepted and the process repeats from step 6.
 5. If the second bid is no longer valid, it is rejected and the bidder is shown: "Someone just beat you to it! You've been outbid."
- **6b. True simultaneous arrival (identical timestamp, unresolvable order):**
 1. System arbitrarily accepts one bid.
 2. The other bidder is notified their bid was rejected and prompted to increase their bid.

Special Requirements:

- Backend timestamp precision: milliseconds
- Frontend timer/display precision: seconds
- This use case is the primary target for TLA+ formal modeling (race conditions, timer extension, no-self-outbid invariant) and mutation testing (the $>/\geq$ bid-comparison operator).